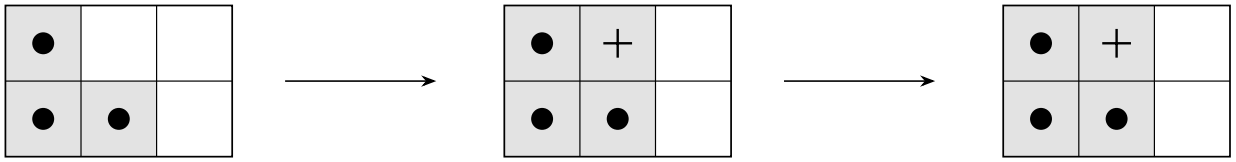
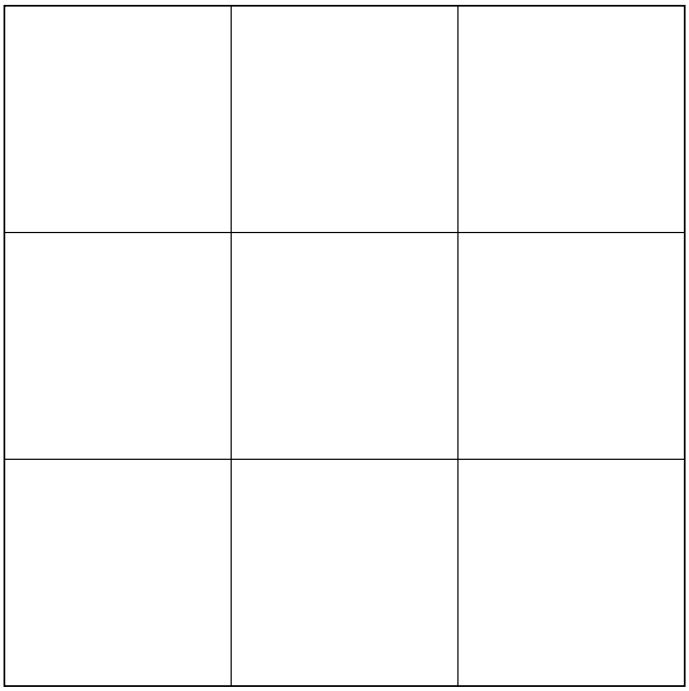
Week 98 / Growing from a few seeds / Grades 2–3

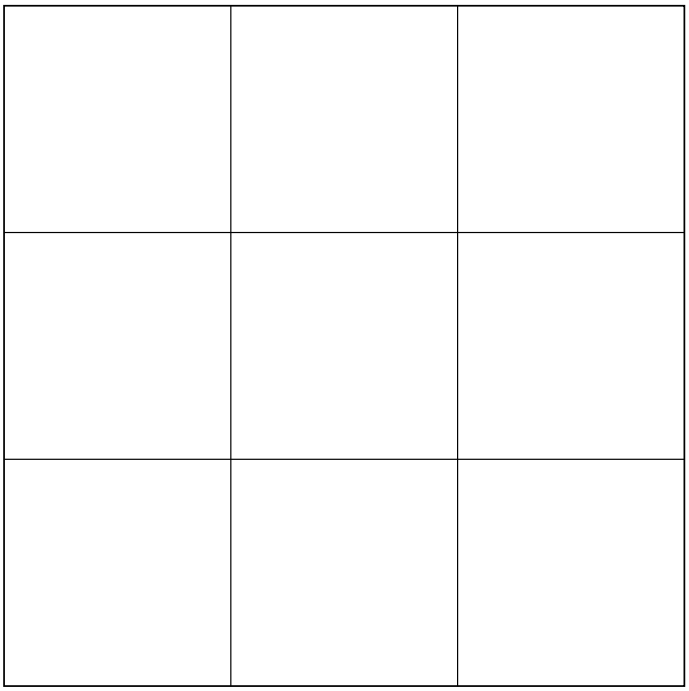
Seeds are the squares filled at the start. Put counters on them. Add a counter to an empty square when at least two squares sharing its sides are filled. Your partner points to two filled neighbors before each addition. Counters stay put. Stop when no more can be added. Diagonal touches do not count. There are no squares outside the board.

Mark seeds with • and added squares with $+$.

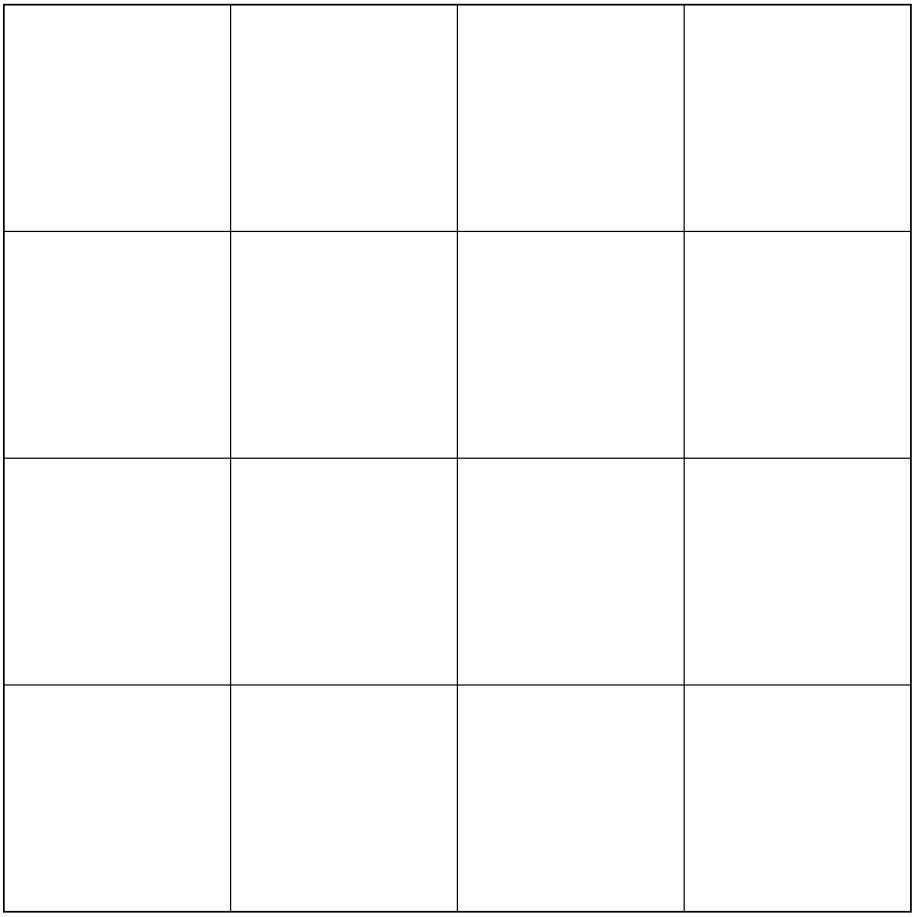


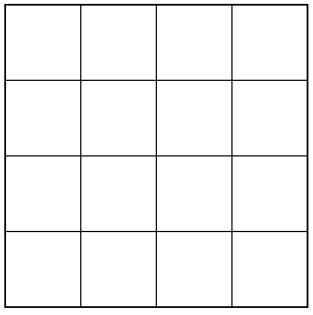
Problem 1: Choose exactly three seeds on each board. Make one start that fills the whole board and one that gets stuck with empty squares.

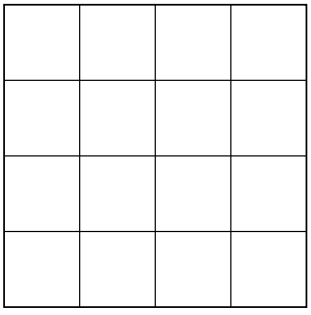


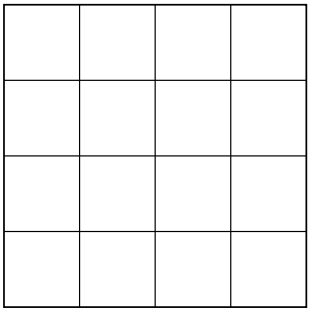


Problem 2: What is the smallest number of seeds you can use to fill a 4-by-4 board? Find three different starts using your best number.



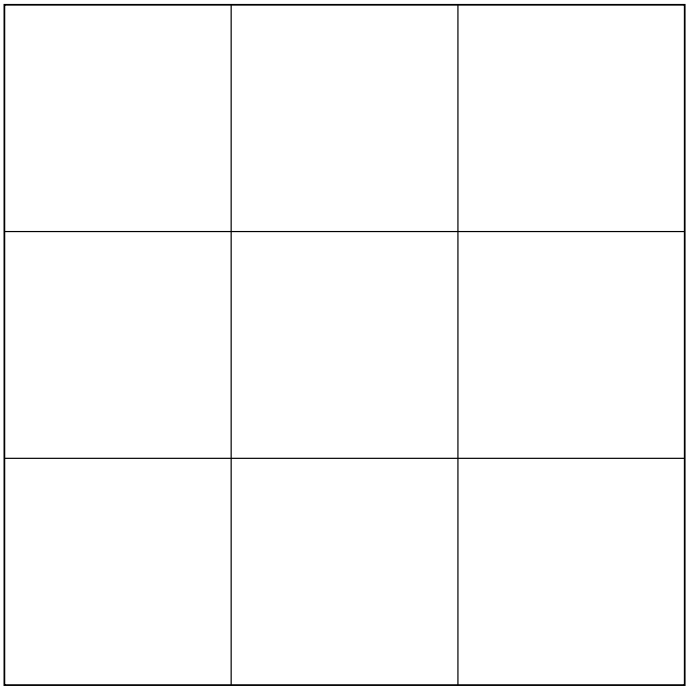




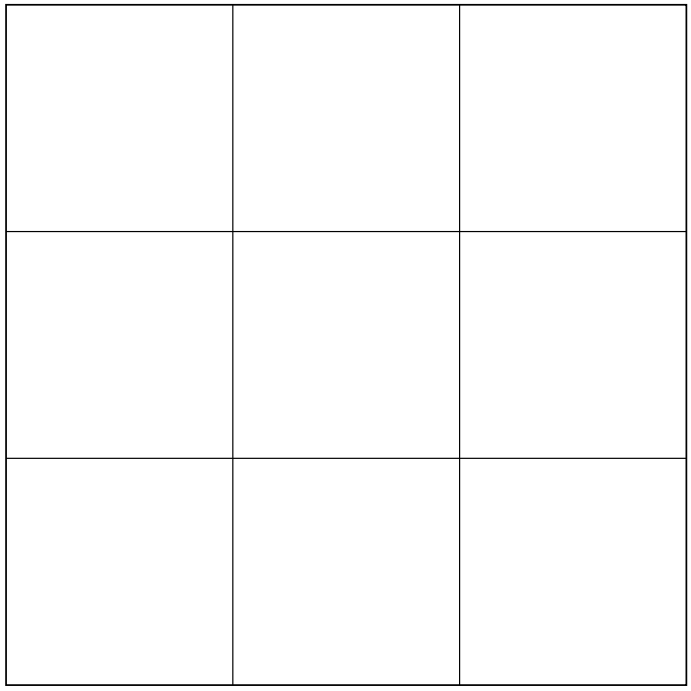


Problem 3: Choose exactly three seeds on each board. Make the finishes have 3, 4, 6 and 9 filled squares. Can three seeds finish with any other number of filled squares?

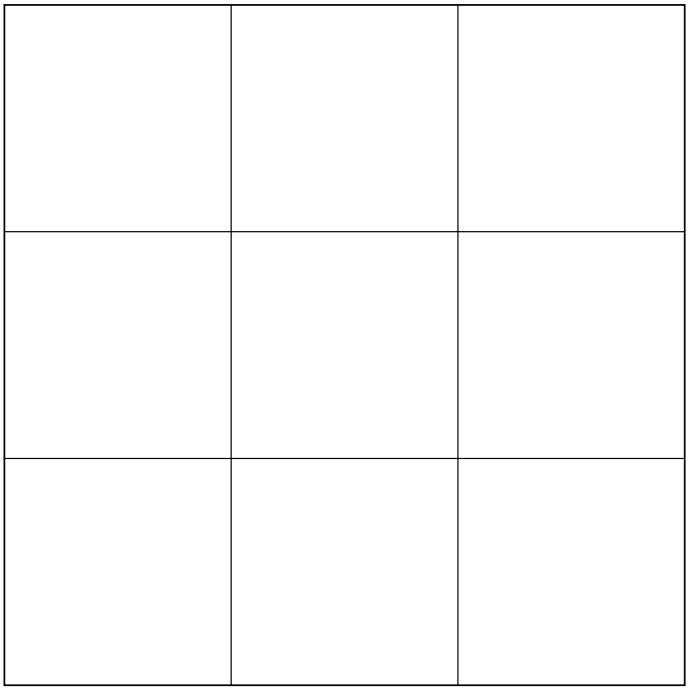
3 filled squares



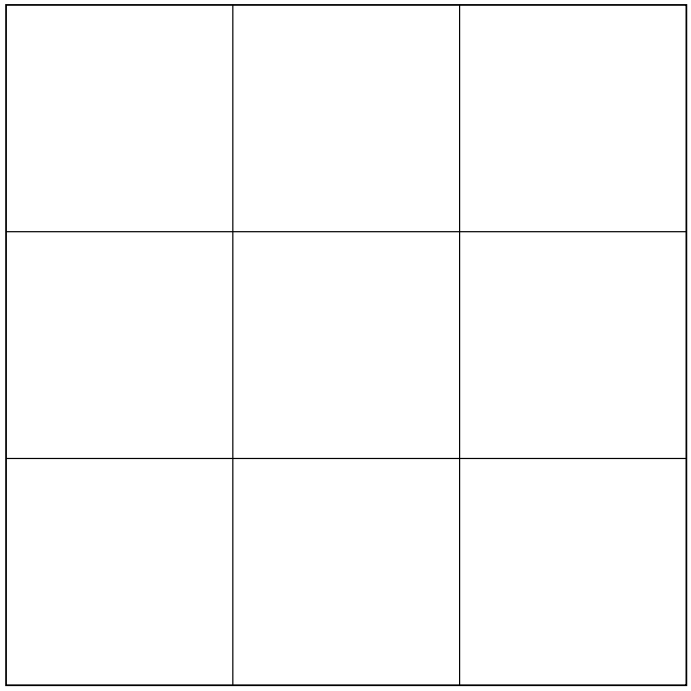
4 filled squares



6 filled squares

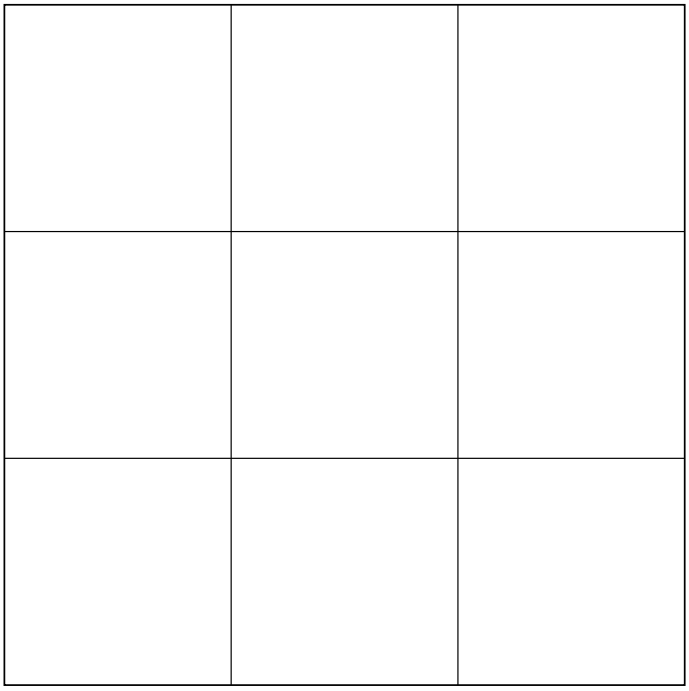


9 filled squares

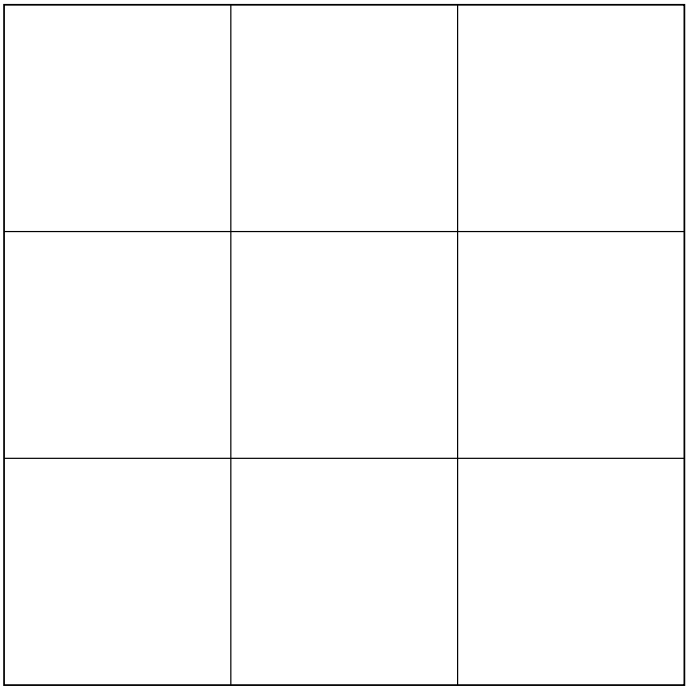


Problem 4: On each pair of boards, use the same three seeds and grow them in different orders when possible. Number the added squares in the order you add them. Can the two finished boards have different filled squares? Test several starts.

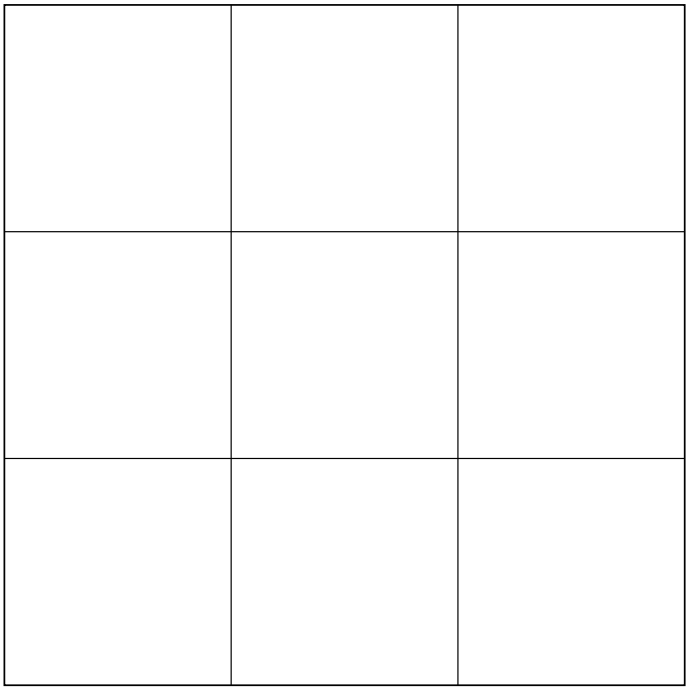
First finish



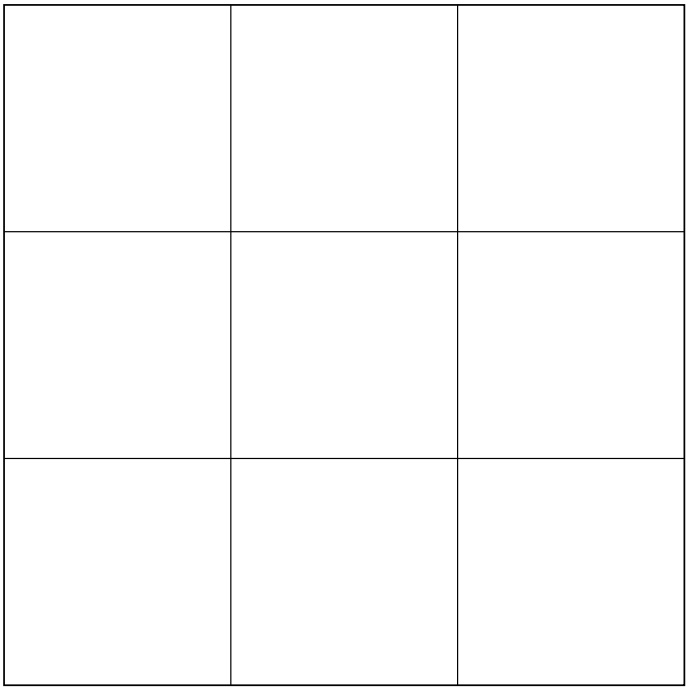
Second finish



First finish



Second finish



Problem 5: Grow the four-seed start. Then test all four ways to leave out one seed and start again. Is every seed needed? Is four the fewest seeds that can fill the board?

